Outputs

1.

```
(myenv) jathin@Victus:~/Documents/ml/week0$ python task1.py
  City1 City2  avrg_fare  Dist  Avrg_weekly_passengers  ... mkt_share_h  avrg_h  low_price_air mkt_share_l  avrg
_l
0   CAK   ATL     114.47   528                  424.56  ...       70.19  111.03             FL       70.19  111.
03
1   CAK   MCO     122.47   860                  276.84  ...       75.10  123.09             DL       17.23  118.
94
2   ALB   ATL     214.42   852                  215.76  ...       78.89  223.98             CO        2.77  167.
12
3   ALB   BWI      69.40   288                  606.84  ...       96.97   68.86             WN       96.97   68.
86
4   ALB   ORD     158.13   723                  313.04  ...       39.79  161.36             WN       15.34  145.
42

[5 rows x 11 columns]
   avrg_fare  Dist  Avrg_weekly_passengers Mkt_leading_air  ...  City2_TLH  City2_TPA City2_TUL  City2_TUS
0     114.47   528                  424.56             FL  ...          0          0         0          0
1     122.47   860                  276.84             FL  ...          0          0         0          0
2     214.42   852                  215.76             DL  ...          0          0         0          0
3      69.40   288                  606.84             WN  ...          0          0         0          0
4     158.13   723                  313.04             UA  ...          0          0         0          0

[5 rows x 184 columns]
```

2.

```
(myenv) jathin@Victus:~/Documents/ml/week0$ python task2.py
1000
1000
```

(No rows with missing values)

3.

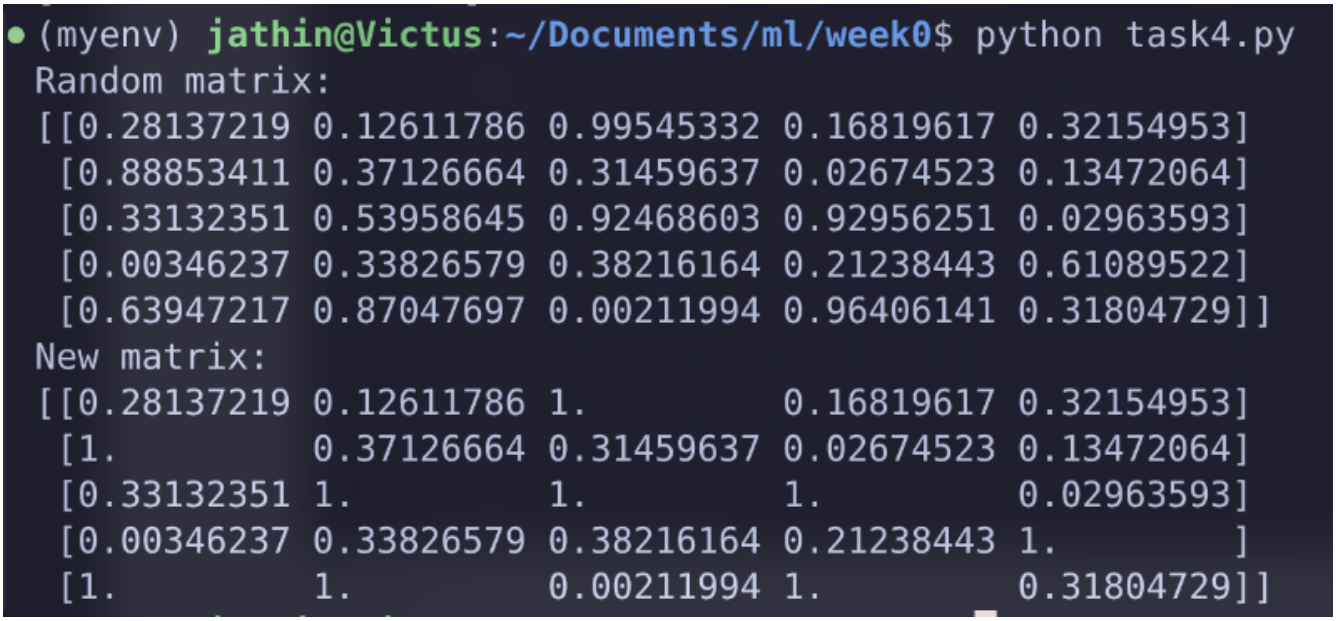
```
(myenv) jathin@Victus:~/Documents/ml/week0$ python task3.py
     City1 City2  avrg_fare  Dist  ...  avrg_h low_price_air  mkt_share_l  avrg_l
184    BOS   CVG     249.21   752  ...  259.41            CO         1.81  114.64
799    BNA   RDU      94.97   443  ...   92.06            WN        88.37   92.06
427    DFW   TPA     194.21   929  ...  196.35            DL        32.39  191.93
364    CMH   BDL     131.81   550  ...  153.81            US        10.21  102.58
589    IAH   BNA     141.06   670  ...  134.31            WN        63.33  134.31

[5 rows x 11 columns] 800
   City1 City2  avrg_fare  Dist  Avrg_weekly_passengers  ... mkt_share_h  avrg_h  low_price_air mkt_share_l  avr
g_l
1    CAK   MCO     122.47   860                  276.84  ...       75.10  123.09             DL       17.23  118
.94
11   ABQ   BWI     173.56  1670                  312.39  ...       49.16  180.49             AA       14.37  161
.39
16   ABQ   LAS     114.24   487                  620.86  ...       93.92  113.82             WN       93.92  113
.82
17   ABQ   LAX     132.29   677                  655.00  ...       89.46  130.44             WN       89.46  130
.44
23   ABQ   PDX     163.63  1111                  222.93  ...       57.24  167.46             HP       20.77  155
.82

[5 rows x 11 columns] 200
```

4.

Outputs



5.

fixed acidity;"volatile acidity";"citric acid";"residual sugar";"chlorides";"free sulfur dioxide";"total sulfur dioxide";"density";"pH";"sulphates";"alcohol";"quality"
0 7;0.27;0.36;20.7;0.045;45;170;1.001;3;0.45;8.8;6
1 6.3;0.3;0.34;1.6;0.049;14;132;0.994;3.3;0.49;9...
2 8.1;0.28;0.4;6.9;0.05;30;97;0.9951;3.26;0.44;1...
3 7.2;0.23;0.32;8.5;0.058;47;186;0.9956;3.19;0.4...
4 7.2;0.23;0.32;8.5;0.058;47;186;0.9956;3.19;0.4...
... ...
4893 6.2;0.21;0.29;1.6;0.039;24;92;0.99114;3.27;0.5...
4894 6.6;0.32;0.36;8;0.047;57;168;0.9949;3.15;0.46;...
4895 6.5;0.24;0.19;1.2;0.041;30;111;0.99254;2.99;0....
4896 5.5;0.29;0.3;1.1;0.022;20;110;0.98869;3.34;0.3...
4897 6;0.21;0.38;0.8;0.02;22;98;0.98941;3.26;0.32;1...

[4898 rows x 1 columns]

fixed acidity;"volatile acidity";"citric acid";"residual sugar";"chlorides";"free sulfur dioxide";"totalsulfur dioxide";"density";"pH";"sulphates";"alcohol";"quality"
count 4898
unique 3961
top 7;0.15;0.28;14.7;0.051;29;149;0.99792;2.96;0.3...
freq 8

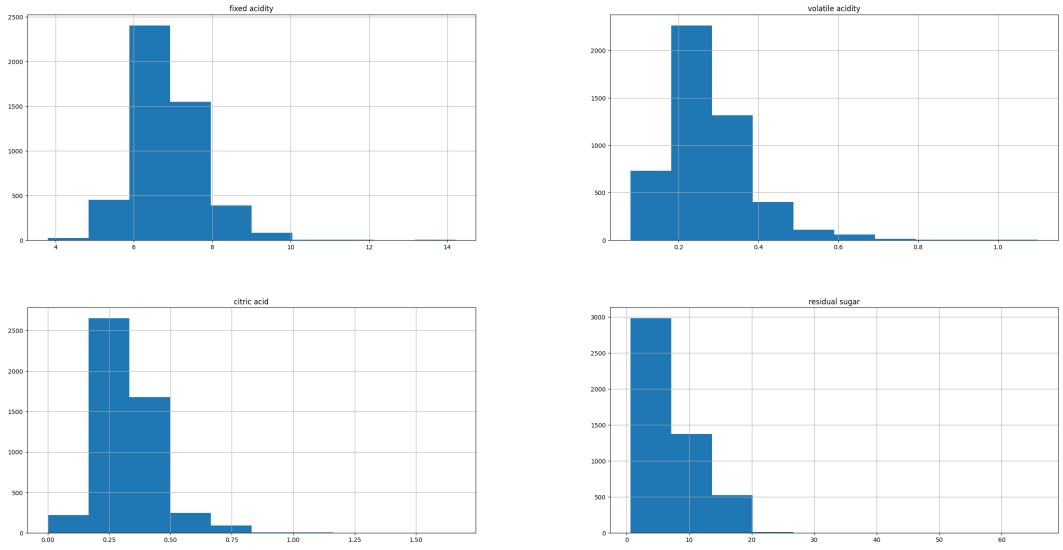
fixed acidity;"volatile acidity";"citric acid";"residual sugar";"chlorides";"free sulfur dioxide";"total sulfur dioxide";"density";"pH";"sulphates";"alcohol";"quality"
0 7;0.27;0.36;20.7;0.045;45;170;1.001;3;0.45;8.8;6
1 6.3;0.3;0.34;1.6;0.049;14;132;0.994;3.3;0.49;9...
2 8.1;0.28;0.4;6.9;0.05;30;97;0.9951;3.26;0.44;1...
3 7.2;0.23;0.32;8.5;0.058;47;186;0.9956;3.19;0.4...
4 7.2;0.23;0.32;8.5;0.058;47;186;0.9956;3.19;0.4...
... ...

Outputs

4893 6.2;0.21;0.29;1.6;0.039;24;92;0.99114;3.27;0.5...
4894 6.6;0.32;0.36;8;0.047;57;168;0.9949;3.15;0.46;...
4895 6.5;0.24;0.19;1.2;0.041;30;111;0.99254;2.99;0....
4896 5.5;0.29;0.3;1.1;0.022;20;110;0.98869;3.34;0.3...
4897 6;0.21;0.38;0.8;0.02;22;98;0.98941;3.26;0.32;1...

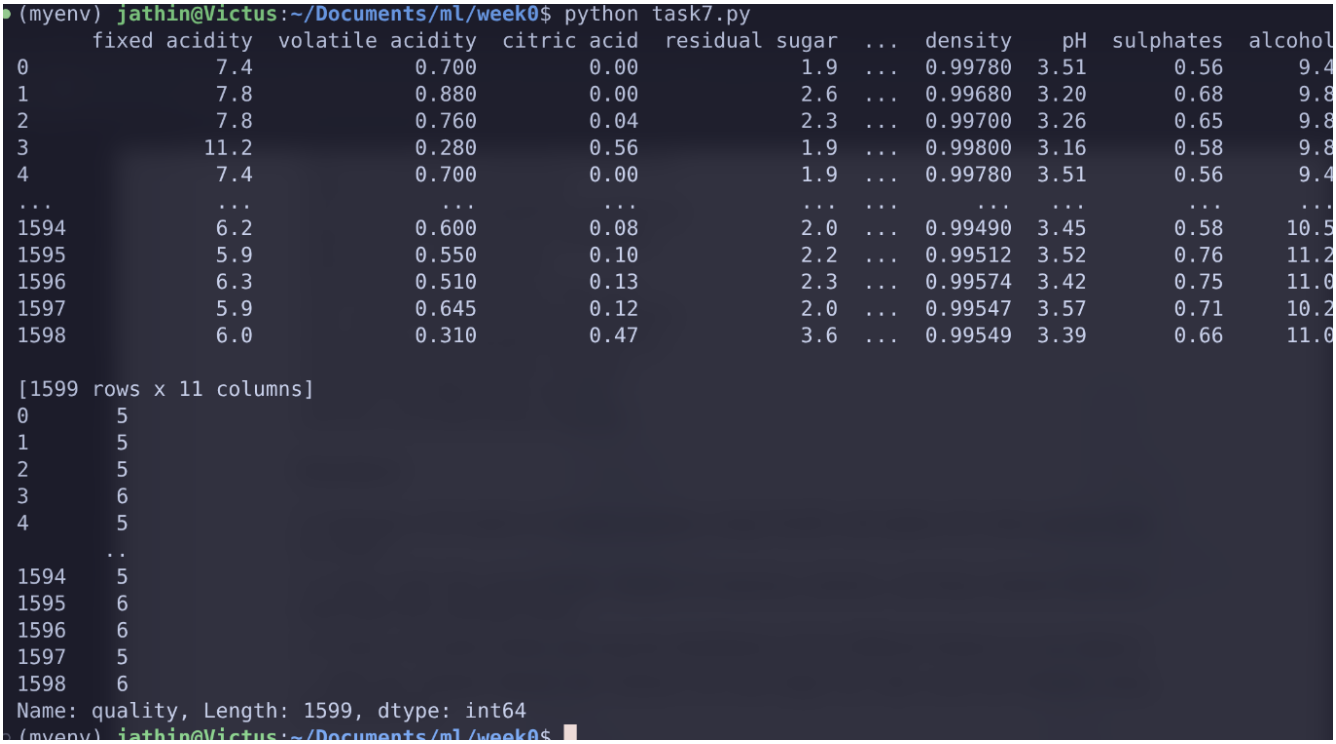
[4898 rows x 1 columns]

6.



7.

Outputs



```
(myenv) jathin@Victus:~/Documents/ml/week0$ python task7.py
      fixed acidity  volatile acidity  citric acid  residual sugar  ...  density    pH  sulphates  alcohol
0               7.4             0.700         0.00             1.9  ...  0.99780  3.51       0.56      9.4
1               7.8             0.880         0.00             2.6  ...  0.99680  3.20       0.68      9.8
2               7.8             0.760         0.04             2.3  ...  0.99700  3.26       0.65      9.8
3              11.2             0.280         0.56             1.9  ...  0.99800  3.16       0.58      9.8
4               7.4             0.700         0.00             1.9  ...  0.99780  3.51       0.56      9.4
...             ...               ...          ...             ...  ...      ...   ...        ...      ...
1594            6.2             0.600         0.08             2.0  ...  0.99490  3.45       0.58     10.5
1595            5.9             0.550         0.10             2.2  ...  0.99512  3.52       0.76     11.2
1596            6.3             0.510         0.13             2.3  ...  0.99574  3.42       0.75     11.0
1597            5.9             0.645         0.12             2.0  ...  0.99547  3.57       0.71     10.2
1598            6.0             0.310         0.47             3.6  ...  0.99549  3.39       0.66     11.0

[1599 rows x 11 columns]
0       5
1       5
2       5
3       6
4       5
       ..
1594    5
1595    6
1596    6
1597    5
1598    6
Name: quality, Length: 1599, dtype: int64
(myenv) jathin@Victus:~/Documents/ml/week0$
```